

EngageNet

Progress Update - August 26, 2026

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The Short Version

What happened since May

The model was built, it trained, and it produced numbers - but the numbers were bad.

I went looking for *why*, and found six problems. Two of them are not opinions: they are arithmetic. The model literally could not do what we designed it to do.

**All six are fixed, the paper is rewritten, and it has now been retrained.
Best validation CCC went from 0.0227 to 0.4228.**

Today's Agenda

1. **Recap** - what the task is, what the model does
2. **Where we stood** - the numbers that started this
3. **The map** - a small theorem telling us where to look
4. **Problem 1** - the model had no memory
5. **Problem 2** - the modalities never talked to each other
6. **Problem 3** - we predicted the wrong thing
7. **Problem 4** - the loss function had three bugs
8. **Results** - what the retrained model actually does
9. **What is still open** - including one thing we got wrong

PART 0: What Is The Task?

The setup

Two people talk over a video link. One is the **expert** (explains something they love), one is the **novice** (listens, asks questions).

Human annotators watch the recording and move a slider to say *how engaged is this person, right now*.

Our job

For **every frame** of video, and for **each of the two people**, predict that slider value.

$$y_t \in [0, 1], \quad 25 \text{ frames per second}$$

A 10-minute session is $\sim 15,000$ numbers per person. We predict all of them.

Recap - The Data We Get

Four feature types $\mathcal{F}$

- ▶ eGeMAPS v2 - acoustic descriptors
- ▶ W2v-BERT 2.0 - learned speech embedding
- ▶ OpenFace2 - facial action units
- ▶ OpenPose - body keypoints

Two roles $\mathcal{R}$

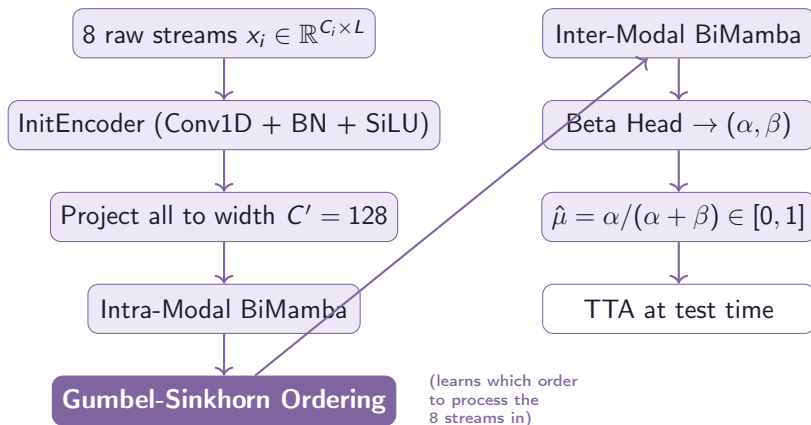
Separate camera and microphone per person, so **every feature type arrives twice**:

$$M = |\mathcal{F}| \times |\mathcal{R}| = 4 \times 2 = 8$$

8 parallel streams into the network.

Remember this $M = 8$. Half of today's talk is about what the model does with these 8 streams.

Recap - Architecture (as of May)



Recap - What A State-Space Model Actually Does

One line of maths, and it is the whole idea

$$\xi_t = \bar{A}\xi_{t-1} + \bar{B}_t u_t$$

ξ_t is the model's **memory** at step t . Each step it keeps a fraction $\bar{A}$ of what it remembered, and adds in the new input u_t .

Why this shape

$\bar{A} \in (0, 1)$, so old information fades smoothly instead of blowing up. Nice and stable.

Why it is fast

Looks sequential, but the update is *affine*, so we can compute all L steps with an associative scan in $O(\log L)$ depth instead of $O(L)$.

Keep $\bar{A}$ in mind. Problems 1 and 2 are both about this one number.

Recap - Why Beta Instead Of A Plain Number

The model outputs a distribution

Not one number, but two: (α, β) , defining

$$p(y \mid x) = \text{Beta}(\alpha, \beta), \quad y \in [0, 1]$$

Prediction is the mean $\hat{\mu} = \frac{\alpha}{\alpha + \beta}$.

Why bother

The *spread* tells us how sure the model is:

$$\text{Var}[y] = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Test-time adaptation uses this to pick which unlabelled frames are safe to learn from.

Beta lives on $[0, 1]$ - exactly the range of engagement. A Gaussian would put probability on -0.3 and 1.4 , which are not things.

PART 1: Where We Stood

What training gave us

- ▶ Best validation CCC: **0.0227**
- ▶ Reached at epoch 2, then flat
- ▶ Correlation $\rho \approx 0.229$
- ▶ 7.2M parameters

What the leaderboard looks like

Official baseline	0.400
2025 winner	0.699
Us	0.02

This is not “needs tuning”. This is “something is structurally wrong”.
So instead of tuning, I went looking for what.

PART 2: A Small Theorem That Told Us Where To Look

Split CCC into two factors

Let $\varphi = \sigma_{\hat{y}}/\sigma_y$ (are the predictions as spread out as the truth?) and $\omega = (\mu_{\hat{y}} - \mu_y)/\sqrt{\sigma_{\hat{y}}\sigma_y}$ (are they centred right?). Then

$$\text{CCC} = \rho \cdot C_b, \quad C_b = \frac{2}{\varphi + \varphi^{-1} + \omega^2}$$

Proposition

$\text{CCC} \leq \rho$, with equality only when $\sigma_{\hat{y}} = \sigma_y$ and $\mu_{\hat{y}} = \mu_y$.

Proof. $\varphi + \varphi^{-1} \geq 2$ for any $\varphi > 0$ (AM-GM), and $\omega^2 \geq 0$. So the denominator of C_b is at least 2, giving $C_b \leq 1$. $\square$

Translation: correlation is a hard ceiling. Rescaling the output can never beat it.

What The Theorem Told Us

Our numbers

$$\rho = 0.229, \quad \text{CCC} = 0.21$$

$$\implies C_b = \frac{0.21}{0.229} \approx 0.92$$

We were already **92% calibrated**.

So what does that mean

Fiddling with output scaling could buy us at most $0.229 - 0.21 = 0.02$.

Every remaining point has to come from ρ - from the model getting the *ordering* of frames right.

The model could not tell engaged moments from bored ones.
Everything that follows is about *why not*.

PROBLEM 1: The Model Had A Goldfish Memory

How long does an SSM remember?

Stop feeding it input. Then $\xi_{t_0+k} = \bar{A}^k \xi_{t_0}$: memory decays geometrically. The standard measure is how long until it drops to e^{-1} :

$$\bar{A}^{\tau_{\text{eff}}} = e^{-1} \implies \tau_{\text{eff}} = \frac{-1}{\ln \bar{A}} = \frac{1}{\Delta |A|}$$

Now plug in what we actually initialised

We set $A_{\log} = 0$ and $s_{\Delta} = 0$. That gives $|A| = e^0 = 1$ and $\Delta = \text{softplus}(0) = \ln 2 \approx 0.693$, so

$$\tau_{\text{eff}} = \frac{1}{0.693} \approx 1.44 \text{ frames} = 58 \text{ ms}$$

Problem 1 - Why 58 ms Is A Disaster

The mismatch

Engagement changes over **seconds**. Our model could see **58 milliseconds**.

That is about one and a half frames. It is like judging whether someone is enjoying a film by looking at a single blink.

Reference Mamba initialises Δ between 10^{-3} and 10^{-1} . Ours was 0.693 - **seven times larger than their maximum**.

And it gets worse

$A_{\log} = 0$ means *every one* of the $N = 16$ state coordinates got the **same** decay rate.

We paid for 16 memories at different timescales and received 16 identical copies of one very short memory.

Problem 1 - The Fix

S4D-Real initialisation

Give each state coordinate its own decay rate, spread geometrically:

$$A_{\log}[d, n] = \ln(n + 1) \implies a_{d,n} = -(n + 1), \quad n = 0, \dots, N - 1$$

and spread Δ log-uniformly over $[10^{-3}, 10^{-1}]$ by inverting the softplus, $s_{\Delta} = \ln(e^{\Delta} - 1)$.

Before

$|A|$: one value

τ_{eff} : 1.4 frames (58 ms)

After (measured)

$|A|$: 16 distinct values, $1 \rightarrow 16$

τ_{eff} : $10 \rightarrow 1000$ frames

(0.4 to 40 seconds)

Two lines of code. The model can now choose its own memory span per channel.

PROBLEM 2: The Modalities Never Talked To Each Other

The whole point of the fusion stage

Take the 8 processed streams and let them share information - so the model can notice “the face says bored *but* the voice says excited”.

What we found

We had **two different architectures**: one described in the paper, one actually running in the code.

They were different from each other, and both were broken - in different ways.

This is the embarrassing one. It is also the one with the biggest payoff.

Problem 2a - What The Paper Described

Lay the 8 streams end to end, scan along that line

Each stream is $C' = 128$ channels wide, so the line is $MC' = 1024$ positions long and **each modality occupies a block of 128**.

The telephone game problem

To get from one modality to the next, information must survive $C' = 128$ steps. Each step keeps a fraction $\bar{A}$:

$$\bar{A}^{C'} = 0.5^{128} \approx 3 \times 10^{-39}$$

The smallest number float32 can represent is $\approx 1.2 \times 10^{-38}$.

The message does not arrive faint. It arrives as exactly zero.

Like whispering down a line of 128 people where each one halves the volume.

Problem 2b - What The Code Actually Did

Concatenate the modalities as *features*, scan along time

Different design. Here modalities *do* mix - through a big dense layer $W \in \mathbb{R}^{MC' \times MC'}$ that touches everything at once. So fusion happens.

But two things go wrong.

The ordering did nothing

Our whole Gumbel-Sinkhorn machinery learns a permutation P . But

$$W(P \otimes I_{C'}) U = W' U, \quad W' := W(P \otimes I_{C'})$$

A dense layer can **absorb** any reordering into its own weights. The permutation was decorative.

It ate the whole model

$D = MC' = 1024$, and the block has five 1024×1024 matrices:

$$5.36\text{M of } 7.20\text{M} = 74\%$$

Three quarters of our parameters, on 31 training sessions.

Problem 2 - The Fix: Make Modality The Sequence

Stop unpacking modalities into channels

Treat each modality's whole C' -vector as **one position**. The sequence is then just the 8 modalities, and we run it once per time step:

$$H^{(t)} = \text{BiMamba}_{\text{modality}}(\tilde{u}_1^{(t)}, \dots, \tilde{u}_M^{(t)}), \quad t = 1, \dots, L'$$

Distance between modalities

Was $C' = 128$ steps $\rightarrow$ now **1 step**.

Attenuation $\bar{A}^{128} \rightarrow \bar{A}$.

Parameter cost

Was $\Theta(M^2 C'^2) \rightarrow$ now $\Theta(C'^2)$.

5.36M $\rightarrow$ 96,640 (a 55x cut).

And now the learned ordering *matters*, because the scan really is sequential over modalities.

PROBLEM 3: We Predicted The Wrong Thing

What we were doing

The challenge asks for **each person's** engagement. We were averaging the two into one number and predicting that:

$$\bar{y}_t = \frac{1}{2}(y_t^e + y_t^n)$$

Why averaging destroys signal

With both annotations having variance σ^2 and correlation r :

$$\text{Var}(\bar{y}) = \frac{1}{4}[\sigma^2 + \sigma^2 + 2r\sigma^2] = \frac{\sigma^2(1+r)}{2}$$

If the two people's engagement moves *oppositely*, averaging flattens the curve towards a straight line.

Problem 3 - How Bad Is It? The Corpus Paper Measured It

Expert-novice correlation r

Language	r	Variance kept
German	-0.19	41%
English	-0.07	47%
French	-0.05	48%
Japanese	+0.51	76%

(Funk et al., 2024, Sec. 5.3.4)

Three of four are negative

On the European sessions we were throwing away **more than half** the signal before training even started.

And CCC divides by the target variance - so this is not recoverable later.

Japanese is the exception; the paper links it to a stronger tendency toward interpersonal harmony.

Problem 3 - The Fix (Easier Than Expected)

No new machinery needed

The two people were **never mixed on the way in** - their recordings enter as separate streams and stay separately indexed the whole way through.

So: attach **one output head per role** instead of one shared head. Each unimodal head is now trained against the person whose stream it reads.

Bonus 1

This is what the challenge actually asked for. We were solving a different problem than the one being scored.

Bonus 2

Gender is per person. With one averaged output, a session where the two differ had *no valid group label* - we were silently dropping those sessions from the fairness metric.

PROBLEM 4: The Loss Function Had Three Bugs

The old loss, in full

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{NLL}}(\alpha, \beta, y) + \lambda_u \sum_{i=1}^M \mathcal{L}_{\text{NLL}}(\alpha_i, \beta_i, y), \quad \lambda_u = 0.5$$

1. It rewarded being *calibrated*, but we are scored on being *correlated*
2. It paid most attention to the frames the model already knew
3. The 8 helper heads outweighed the main head by 4x

Bonus defect: nothing in it ever asked the model to be fair.

Bug 4a - Calibrated Is Not The Same As Correlated

What the likelihood asks

“At *this* frame, is your predicted distribution right?”

One frame at a time. Never compares frames to each other.

What CCC asks

“Across *all* frames, do your predictions go up and down in the same pattern as the truth?”

And by Part 2, that is the only thing that can raise our score.

Fix: add the metric itself to the loss

$$\mathcal{L} += \lambda_c (1 - \widehat{\text{CCC}})$$

Every ingredient of CCC (two means, two variances, one covariance) is a smooth function of the predictions, so we can just differentiate it.

Bug 4b - The Loss Ignored The Hard Frames

Differentiate the Beta log-likelihood w.r.t. the mean

Writing $\mu = \alpha/(\alpha + \beta)$, $\kappa = \alpha + \beta$ (so κ = confidence), and using $\partial\alpha/\partial\mu = \kappa$, $\partial\beta/\partial\mu = -\kappa$, the two $\psi(\alpha+\beta)$ terms cancel and leave

$$\frac{\partial(-\ln p)}{\partial\mu} = \kappa \left[\ln \frac{1-y}{y} + \psi(\mu\kappa) - \psi((1-\mu)\kappa) \right] \propto \kappa$$

Read that again

The correction is multiplied by the model's own confidence. Frames it is *already sure about* produce the biggest updates; uncertain frames produce almost none.

It is a teacher who only helps the students who already understand.

Bug 4b - The Fix, And Why It Needs stop_grad

Re-weight each frame

$$w_t = \text{stop_grad}(\kappa_t^{-\beta_w}), \quad \implies \quad \frac{\partial \mathcal{L}}{\partial \mu} \propto \kappa^{1-\beta_w}$$

$\beta_w = 0$ is the old behaviour, $\beta_w = 1$ removes confidence weighting entirely. We use $\beta_w = 0.5$.

Why the weight must be detached

If gradients could flow through w_t , the model would spot a shortcut: make κ huge, make w_t tiny, and the loss goes down *without predicting anything better*.

`stop_grad` means “compute this number, then treat it as a constant”. It rebalances the update without becoming another thing to game.

Bug 4c - The Helper Heads Were Running The Show

The arithmetic

$M = 8$ helper heads, each weighted $\lambda_u = 0.5$:

$$8 \times 0.5 = 4$$

versus 1 for the main head.

And at test time we **only use the main head**.

The fix

Divide by M :

$$\frac{\lambda_u}{M} \sum_{i=1}^M \mathcal{L}_{\text{NLL}}(\alpha_i, \beta_i, y^{r(i)})$$

Now the helpers contribute 0.5 total, not 4, and adding more modalities does not silently change the balance.

80% of the gradient was coming from parts of the network we throw away at evaluation.

Bug 4d - We Measured Fairness But Never Asked For It

CDD is differentiable, so we can train against it

Split the batch into n_b quantile bins of the ground truth. Inside each bin, compare the two groups' average predictions:

$$\mathcal{L}_{\text{fair}} = \frac{\sum_{b=1}^{n_b} |\mathcal{B}_b| (\bar{\hat{\mu}}_b^A - \bar{\hat{\mu}}_b^B)^2}{\sum_{b=1}^{n_b} |\mathcal{B}_b|}$$

Squaring removes the sign, so it is smallest exactly when both groups get the same prediction at the same true engagement.

Honest caveat

This term is **zero** on a batch containing only one group. Gender is a session-level property, so most of our batches are single-group.

It may need window-level shuffling to do anything. Landing it last, and measuring first.

The New Loss, All Together

$$\mathcal{L} = \underbrace{\frac{1}{|\mathcal{R}|} \sum_r \mathcal{L}_{\text{NLL}}^w(\alpha^r, \beta^r, y^r)}_{\text{per-frame fit, rebalanced}} + \underbrace{\lambda_c(1 - \widehat{\text{CCC}})}_{\text{ordering}} + \underbrace{\frac{\lambda_u}{M} \sum_i \mathcal{L}_{\text{NLL}}^w(\alpha_i, \beta_i, y^{r(i)})}_{\text{per-modality}} + \underbrace{\lambda_f \mathcal{L}_{\text{fair}}}_{\text{fairness}}$$

Term	Asks the model to...	Weight
Beta NLL (reweighted)	get each frame's distribution right	$1, \beta_w = 0.5$
$1 - \widehat{\text{CCC}}$	rank frames correctly	$\lambda_c = 1.0$
Per-modality NLL	keep every stream individually useful	$\lambda_u/M = 0.0625$
Squared CDD	treat groups equally at equal truth	$\lambda_f = 0.05$

What Changed, In Numbers

	Before	After
Memory span τ_{eff}	1.4 frames (58 ms)	10–1000 frames (0.4–40 s)
Distinct decay rates	1	16
Cross-modal distance	$C' = 128$ steps	1 step
Fusion block parameters	5,360,000 (74%)	96,640 (5.0%)
Total parameters	7,197,898	2,064,844
Output heads	1 shared	2 (one per person)
Loss terms	2	4
Permutation actually used	no	yes
Best validation CCC	0.0227	0.4228

An $18.6\times$ improvement. Same data, same optimiser, same window length.

PART 3: The Retrained Run

Run summary

Wall time	36 min 38 s
XLA compile	6 min 35 s
Per epoch	~73 s
Epochs run	25 of 50
Early stop	patience 10
Best epoch	15

Best epoch (15)

val CCC	0.4228
CDD _G	+0.0053
CDD _L (Chinese)	-0.027
CDD _L (Japanese)	+0.009

Our own success criterion was $\text{CDD} \leq 0.05$.

Fairness is comfortably inside target. Accuracy is the part still moving.

The Run Was Unstable - And Trained On The Wrong Corpus

Validation CCC bounced hard

0.10 $\rightarrow$ 0.25 $\rightarrow$ 0.19 $\rightarrow$ 0.09 $\rightarrow$ 0.24 $\rightarrow$ 0.24 $\rightarrow$
0.37 $\rightarrow$ 0.38 $\rightarrow$ 0.25 $\rightarrow$ 0.21 $\rightarrow$ 0.08 $\rightarrow$ 0.08 $\rightarrow$
0.34 $\rightarrow$ 0.33 $\rightarrow$ 0.42 $\rightarrow$ 0.29...

Meanwhile training loss fell *monotonically*,
0.058 $\rightarrow$ -0.503. Smooth loss, thrashing
metric.

So 0.4228 is an *in-domain* number on Asian data. Cross-cultural transfer - the entire point of the TTA design - remains untested.

And the corpus problem

The log shows only NoXi+J/train and NoXi+J/val. Languages: Japanese and Chinese only.

NoXi was never loaded. The model has never seen a European session.

Done ✓

- ▶ All six fixes in code
- ▶ Paper rewritten to match
- ▶ Retrained end to end
- ▶ CCC 0.0227 → 0.4228
- ▶ CDD inside target on both axes

Caveats on that number

- ▶ NoXi+J only - no European data
- ▶ Val split is 10 sessions; noisy
- ▶ Not comparable to the 0.400 baseline, which is a mean over four *test* sets
- ▶ Overfits after epoch 15

The fixes clearly worked. Whether the model is competitive is a different question, and one we cannot answer yet.

Blocking

- ▶ **MPIIGroupInteraction** - we do not have it. It is 1 of the 4 test sets, so we cap at 3/4 of the score
- ▶ TTA still never run end-to-end - and it is our headline contribution

Known gaps

- ▶ No per-role or ρ logging - cannot tell if the two heads collapsed
- ▶ MC dropout (paper describes it, code has none)
- ▶ Hungarian hard permutation
- ▶ Score matrix missing $1/\sqrt{d_s}$

Next, in order

1. Train on **NoXi**
2. Log per-role CCC and ρ
3. Stabilise: $\eta = 2 \times 10^{-4}$, patience 5
4. Run TTA NoXi → NoXi+J
5. Ablate the four loss terms

What I Would Like From You

Questions I cannot answer alone

- ▶ **MPIIGroupInteraction** - does anyone have a contact who can speed this up? Without it we lose a quarter of the score by default.
- ▶ **Training strategy** - one model on the union of corpora, or separate models per corpus? The challenge requires us to declare which. Now that a run costs about an hour we can simply test both, but I would like agreement on the criterion first.
- ▶ **Val split size** - 10 sessions gives a CCC estimate that swings by 0.3 between epochs. Do we accept that noise, or move to cross-validation?

Questions?